

Comparative Study of Classification Algorithms for Diabetes Prediction

Emanuele Smisi

Giacomo Aroni

November 2025

1 Introduction and Motivation

The objective of this project is to develop a practical understanding of the main supervised learning algorithms. This goal is pursued through the implementation, analysis, and comparison of the performance of different classification models on a real-world dataset.

For this study, a **binary classification** task was chosen, applied to a problem of significant real-world relevance: diabetes prediction. The motivation for this choice lies not only in the intrinsic importance of the medical problem but also in the specific challenges that such a dataset presents.

2 Dataset Description

The dataset used is the "*Diabetes Prediction Dataset*" (iammustafatz), sourced from the [Kaggle](#) platform.

The dataset is indeed imbalanced, placing a critical emphasis on the need to correctly manage the trade-off between False Positives and False Negatives, where a False Negative (an unidentified diabetic patient) has a much higher cost.

It consists of approximately **100,000 observations** and 9 total variables. These are divided into 8 independent variables (features), which describe the patient's health status and habits, and 1 dependent variable (target), which indicates the presence or absence of diabetes.

The dataset exhibits a significant class imbalance, with approximately **8.4%** of patients being diabetic and **91.6%** being non-diabetic. This imbalance further justifies the critical emphasis on minimizing False Negatives.

2.1 Variables Description

Table 1: Description of independent (features) and dependent (target) variables.

Variable	Description	Type
gender	Patient's sex	Categorical (Dichotomous)
age	Patient's age	Numerical (Continuous)
hypertension	Presence of hypertension	Categorical (Dichotomous, 0: No, 1: Yes)
heart_disease	Presence of heart disease	Categorical (Dichotomous, 0: No, 1: Yes)
smoking_history	Patient's smoking history	Categorical (Nominal)
bmi	Body Mass Index	Numerical (Continuous)
HbA1c_level	Glycated hemoglobin level in blood	Numerical (Continuous)
blood_glucose_level	Blood glucose level	Numerical (Continuous)
Dependent Variable (Target)		
diabetes	Presence or absence of diabetes	Categorical (Dichotomous, 0: No, 1: Yes)

3 Methodology and Models

In this section, we'll present the preprocessing steps applied to the dataset and the classification models implemented for this study.

3.1 Part 2: Data Preprocessing

The dataset was prepared for training by following several fundamental steps to clean, transform, and structure the data optimally for machine learning algorithms.

3.1.1 Handling Missing or Noisy Data

As a first analysis, the presence of missing or noisy data was checked. The dataset was found to be already clean, with no explicit missing values (*nulls*). However, refinement was performed on two categorical variables to improve their consistency:

- **gender:** 18 instances with the value "Other" were identified; given their uniqueness and very low frequency, these were removed so as not to affect the training.
- **smoking_history:** This variable required an in-depth analysis to simplify and clarify it.
 - The value "ever" (present in 4,004 observations) was considered misleading and difficult for the models to interpret. These rows were therefore removed from the dataset.

- The values "former" and "not current" were considered synonymous, as both indicate ex-smokers. They were therefore grouped into the new single category "**past smoker**".
- The "No Info" value, given its high frequency, was kept as an informative category in its own right, rather than being treated as missing or combined with other categories.

While other techniques like oversampling (e.g., SMOTE) or undersampling exist, the chosen method of applying `class_weight` was favored for its direct integration into the model's loss function, effectively penalizing misclassifications of the minority class without altering the dataset's structure.

3.1.2 Feature Encoding and Scaling

To make the data interpretable by mathematical models, two distinct transformations were applied based on the variable type:

- **Categorical Variables:** Non-numeric variables, such as `gender` and `smoking_history`, were transformed into binary (dummy) variables using the **One-Hot Encoding** technique.
- **Numerical Variables:** Continuous variables, such as `age`, `bmi`, `HbA1c_level`, and `blood_glucose_level`, underwent **Standardization** (using `StandardScaler`) to ensure they all contribute equally to the model training, preventing features with larger numerical ranges from dominating the learning process.

It should be noted that the two variables left out of this process (`hypertension` and `heart_disease`) were already dichotomous by nature (0: No, 1: Yes) and therefore did not require further processing, being ignored by the Transformer.

To apply these diverse processing steps simultaneously and streamline the workflow, a **ColumnTransformer** (referred to as a "preprocessor") was employed. Its purpose is to apply different transformations (Scaling and One-Hot Encoding) to specific subsets of columns in a single, coherent step.

3.1.3 Data Split

The dataset was then split into three distinct sets to train, validate, and test the models rigorously:

- **Training Set (60%):** Used for training the models.

- **Validation Set (20%):** Used for hyperparameter tuning and cross-validation, allowing for unbiased model selection.
- **Test Set (20%):** Used for the final, impartial evaluation of the models' performance on unseen data, ensuring robust generalization assessment.

3.2 Part 3: Model Implementation and Training

The implementation and training of all models were performed in the **Google Colab** environment using key Python libraries such as *scikit-learn* for machine learning algorithms, *Pandas* for data manipulation, and *NumPy* for numerical operations.

3.2.1 Handling the Imbalanced Dataset

The main challenge of this dataset is the significant class imbalance. To address this problem, the `class_weight = 'balanced'` parameter was applied to all models that support it (LR, SVM, DT, and RF) to penalize errors on the minority class and maximize **Recall**. This strategy ensures that the models are more sensitive to identifying diabetic patients, which is critical given the higher cost associated with False Negatives in a medical context.

3.2.2 Selected Models

The following models, as required by the assignment, were trained and evaluated:

- **Gaussian Naïve Bayes (GNB)**
- **Decision Tree (DT)**
- **Random Forest (RF)**
- **Logistic Regression (LR)**
- **Support Vector Machine (SVM)**

3.2.3 Hyperparameter Optimization

To identify the optimal configuration for the models, different strategies were employed, focused on maximizing the **F1-Score**, which is particularly suitable for imbalanced datasets as it balances Precision and Recall.

- **Gaussian Naïve Bayes:** This model has no significant hyperparameters to tune and was fitted directly using its default settings.
- **Decision Tree (DT) & Random Forest (RF):** These models were also optimized via hyperparameter tuning, specifically focusing on parameters such as `max_depth` (maximum depth of the tree), `min_samples_split` (minimum samples required to split an internal node), and for Random Forest, `n_estimators` (number of trees in the forest), to find the best-performing estimators.
- **Logistic Regression:** An exhaustive `GridSearchCV` (`cv=5`) identified `'C': 0.1` as the optimal regularization parameter, controlling the trade-off between model simplicity and training data fit.
- **Support Vector Machine (SVM):** A `GridSearchCV` (`cv=3`) on a subset of 15,000 samples identified `{'C': 0.01, 'kernel': 'linear'}` as optimal. The model was then re-trained on the full dataset with these optimized parameters.

3.2.4 Computational Cost

The computational cost for training (or optimizing) the models was recorded, highlighting significant differences crucial for practical deployment.

Table 2: Comparison of model training/optimization times.

Model	Time (seconds)	Notes
Gaussian Naïve Bayes	0.14 s	Full Training Set (Fit only)
Decision Tree	16.32 s	GridSearchCV (50 fits, Full Set)
Random Forest	78.94 s	GridSearchCV (18 fits, Full Set)
Logistic Regression	3.38 s	GridSearchCV (25 fits, Full Set)
Support Vector Machine	49.65 s	GridSearchCV (15 fits, Fullset but using Subset)

Cost Analysis The **Gaussian Naïve Bayes** is (obviously) by far the fastest model, training in only 0.14 seconds, making it highly efficient for rapid prototyping. The **Logistic Regression** optimization was also extremely fast (3.38 seconds). Conversely, the **RF** proved the most expensive, requiring 78.94 seconds for a search even if pruned, indicating higher computational demands for its hyperparameter tuning.

4 Results and Analysis

This chapter presents and analyzes the results obtained from all five models. The performance was calculated on the **Validation Set** (19,196 samples),

ensuring an unbiased evaluation of model generalization before testing.

4.1 Evaluation Metrics

Table 3 summarizes the key metrics for all implemented models.

Table 3: Comparative table of optimized models (on the Validation Set).

Metric	GNB	DT	RF	LR	SVM
Accuracy	90.68%	91.02%	97.09%	89.03%	88.87%
<i>Metrics for the "Diabetes" (Positive) class</i>					
Precision	0.46	0.48	0.93	0.43	0.42
Recall	0.67	0.86	0.70	0.90	0.90
F1-Score	0.55	0.62	0.80	0.58	0.58
ROC AUC	0.92	0.94	0.97	0.97	0.97

4.2 Comparative Analysis

From the analysis of Table 3, a complex picture emerges, highlighting the nuanced trade-offs between different classification algorithms:

- **Random Forest (RF):** This model is the clear "winner" on paper regarding overall performance. It achieves the highest **Accuracy (97.09%)**, **Precision (0.93)**, and **F1-Score (0.80)**. However, this comes at a massive cost in terms of sensitivity to the minority class: its **Recall is only 0.70**, the worst of the group besides GNB. This means it misses nearly 30% of diabetic patients, which is a critical drawback in a medical context where False Negatives are highly undesirable.
- **Decision Tree (DT):** This model offers a good balance between performance metrics on unseen data for Recall, achieving the second-highest F1-Score (0.62) and a strong Recall of 0.86. However, as shown in its learning curve (Figure 1), it exhibits signs of high variance, which could impact its generalization to completely new data.
- **Logistic Regression (LR) & SVM:** These two linear models perform almost identically, demonstrating robust and consistent behavior. They are the undisputed champions of **Recall (0.90 and 0.90)**, successfully fulfilling our primary goal of finding the most diabetic patients. Their F1-Scores (0.58) are solid for imbalanced data, and their ROC AUC (0.97) is tied for the highest, indicating excellent discriminatory power.

- **Gaussian Naïve Bayes (GNB):** This model is the weakest performer across most metrics. Its Recall is only 0.67, making it largely unsuitable for this critical medical task.

4.2.1 Overfitting Analysis and Learning Curves

The metrics in Table 3 provide a snapshot of performance but can be misleading without analyzing the models' reliability and generalization capabilities. The learning curves (Figures 1, 2, and 3) are therefore essential for a deeper understanding.

- **Random Forest (RF):** This model shows **extreme overfitting**. Its learning curve (Figure 1, right) shows a perfect training F1-score of 1.0, while the validation score is far lower (around 0.85-0.90). This large gap indicates that the model has memorized the training data rather than learning generalizable patterns, rendering it unreliable for unseen data.
- **Decision Tree (DT):** This model's learning curve (Figure 1, left) also shows a clear case of **high variance (overfitting)**, with a large, persistent gap between the training and validation scores. This suggests it struggles to generalize beyond the training set.
- **LR and SVM:** (Figure 2). Their learning curves **converge perfectly**, indicating stable, low-variance models that generalize extremely well to new data. This is a strong indicator of their robustness and reliability.
- **Gaussian Naïve Bayes (GNB):** (Figure 3, left). This curve shows **high bias (underfitting)**. Both scores converge, but at a consistently low F1-Score (≈ 0.53), suggesting the model is too simplistic to capture the underlying patterns in the data effectively.

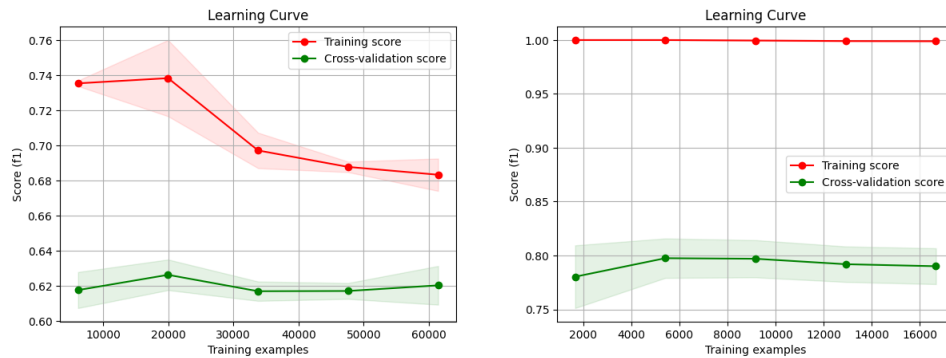


Figure 1: Learning curves for Decision Tree (left) and Random Forest (right). Both models show significant overfitting (high variance).



Figure 2: Learning curves for Logistic Regression (left) and SVM (right). The convergence of the lines indicates excellent generalization.

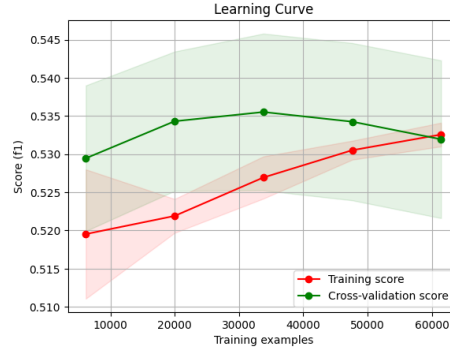


Figure 3: Learning Curve for GNB , showing high bias.

4.2.2 Visualizations: Confusion Matrices and ROC Curves

The confusion matrices (Figures 4, 5, and 6) provide a granular view of the models' predictions and confirm their underlying strategies and biases.

- **LR & SVM:** (Figure 5). These models produce the lowest number of False Negatives (**160** and **154** respectively), which is crucial for our medical task. This outcome directly confirms their high-Recall strategy and suitability for identifying diabetic patients.
- **Random Forest:** (Figure 4, right). This model produces the highest number of False Negatives (**480**) but also the lowest number of False Positives (**79**). This trade-off explains its high Precision (correctly identified positives) but poor Recall (missed actual positives).
- **Decision Tree:** (Figure 4, left). This model strikes a moderate balance between False Negatives and False Positives, with **222** FN and **1501** FP.

- **Gaussian Naïve Bayes:** (Figure 6, left). This model exhibits the second-highest number of FNs (**533**), consistently missing a significant proportion of positive cases, reinforcing its unsuitability for this problem.

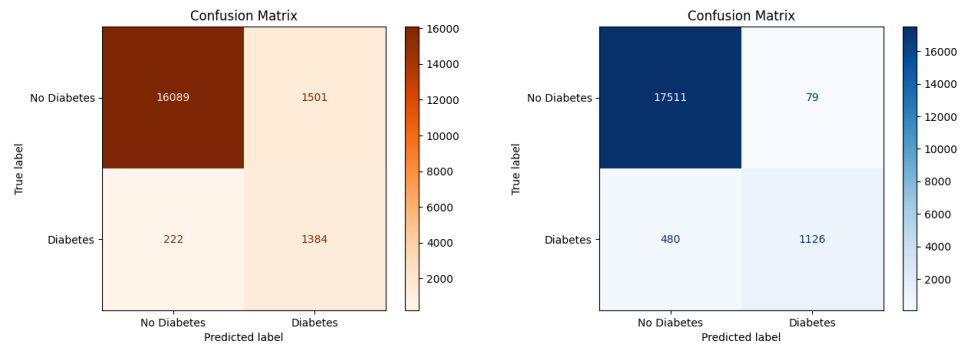


Figure 4: Confusion Matrices for Decision Tree (left) and Random Forest (right).

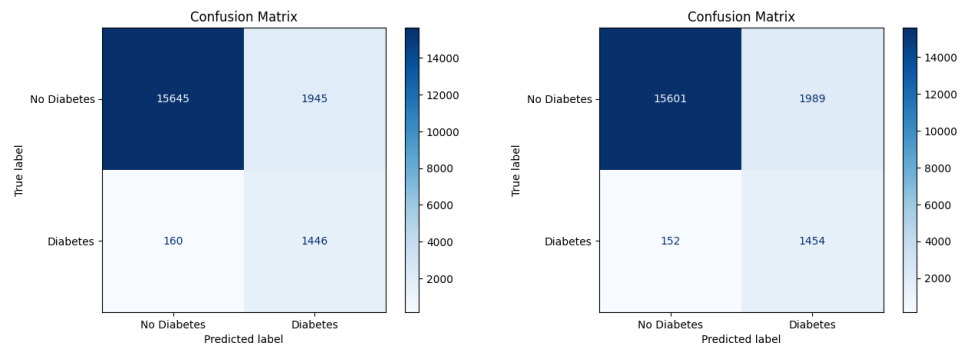


Figure 5: Confusion Matrices for Logistic Regression (left) and SVM (right).

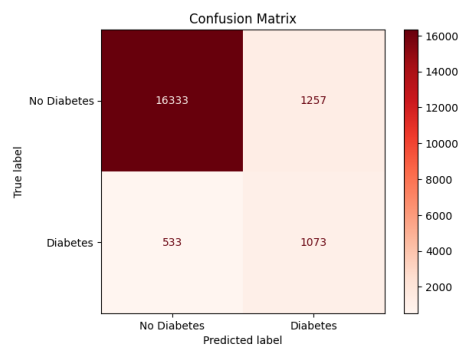


Figure 6: Confusion Matrix for GNB.

The ROC Curves (Figures 7, 8) further confirm that LR, SVM, and RF (all with an AUC of 0.97) are the best discriminators, followed by DT (0.94) and GNB (0.92). High AUC values indicate a strong ability to distinguish between positive and negative classes across various thresholds.

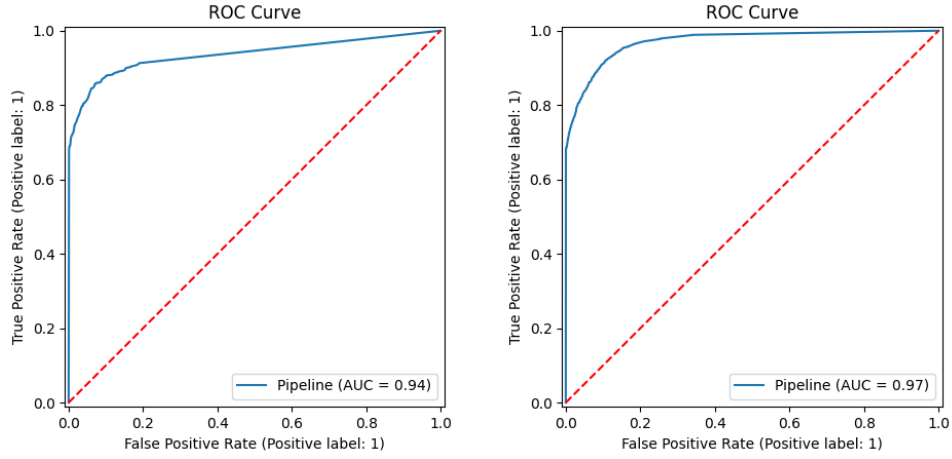


Figure 7: ROC Curves for Decision Tree (left, AUC=0.94) and Random Forest (right, AUC=0.97).

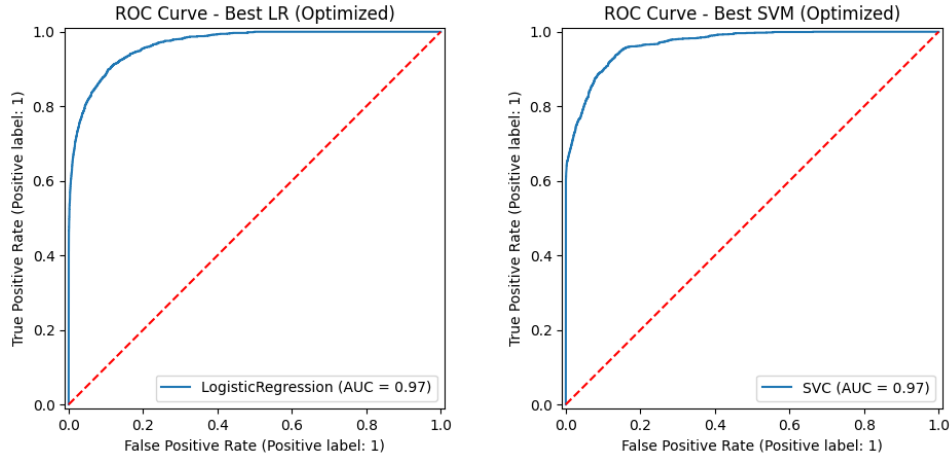


Figure 8: ROC Curves for Logistic Regression (left) and SVM (right). Both models achieve a top-tier AUC of 0.97.

4.2.3 Feature Importance

Finally, the importance of the features was analyzed to understand which predictors contribute most significantly to the models' decisions. We compared the coefficients of the linear models (LR, SVM) with the Gini importance of the tree-based models (DT, RF) (Figures 9 and 10).

All models are in **strong agreement** on the key predictors. `HbA1c_level` and `blood_glucose_level` are, by far, consistently identified as the most important features. `age` and `bmi` are also consistently ranked as highly predictive. This remarkable alignment across all model types (linear, single tree, and ensemble) gives us high confidence in these findings, indicating that these four features are the primary determinants for diabetes prediction in this dataset.

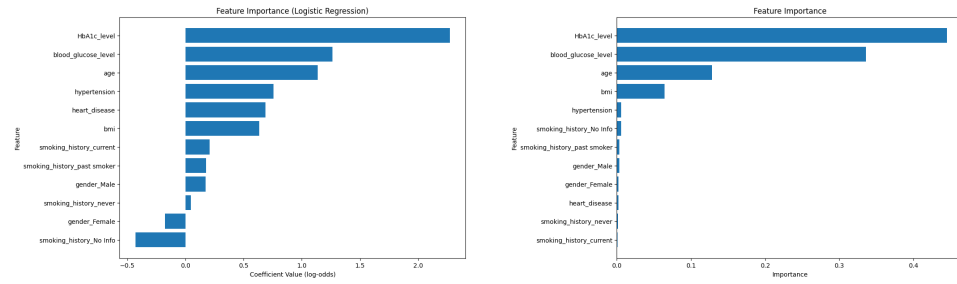


Figure 9: Feature importance from Logistic Regression (left) and Decision Tree (right).

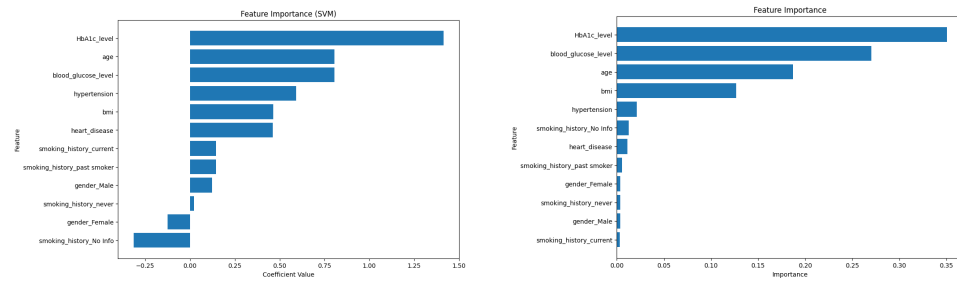


Figure 10: Feature importance from SVM (left) and Random Forest (right). All models agree on the top 4 features.

5 Conclusions

In conclusion, this comparative study of five classification algorithms has yielded a clear and decisive result regarding their suitability for diabetes prediction in an imbalanced dataset.

The non-linear models, **Random Forest (RF)** and **Decision Tree (DT)**, despite achieving the highest on-paper F1-Scores (0.80 and 0.62, respectively), demonstrated significant limitations. Their learning curves reveal that they both suffer from significant **high variance (overfitting)**. The RF model, in particular, showed extreme overfitting (perfect 1.0 training F1-score) and, critically, had the worst Recall (0.70) for this medical task, missing a large portion of actual diabetic patients. These models are therefore unreliable for deployment.

The **Gaussian Naïve Bayes (GNB)** model, while computationally instantaneous (0.14s), suffered from **high bias (underfitting)** and its overall performance (0.67 Recall) was insufficient, rendering it unsuitable for this predictive task.

This thorough analysis leaves the two linear models, **Logistic Regression (LR)** and **Support Vector Machine (SVM)**, as the clear best-in-class solutions. They both provide an excellent balance of performance (F1-Score 0.58) and, most importantly, achieved the highest **Recall** (0.90-0.90), successfully fulfilling the project’s primary goal of minimizing False Negatives in diabetes diagnosis.

Critically, their learning curves confirm they are robust, low-variance models that **generalize extremely well** to unseen data. Given that their performance is statistically identical, but the **Logistic Regression** is significantly more computationally efficient (3.38s for optimization compared to 49.65s for SVM), it stands out as the most practical, reliable, and efficient solution for this specific problem.